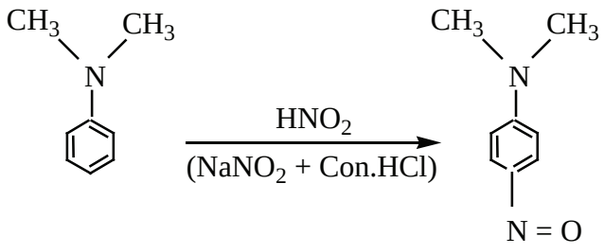
$$(CH_3)_3N + HNO_2 \longrightarrow [(CH_3)_3\overset{+}{N}H]\ NO_2^-$$

trimethylamine — trimethyl ammonium nitrite

ii) Aromatic tertiary amine reacts with nitrous acid at 273K to give p – nitroso compound.



N, N - dimethyl aniline — p - Nitroso N,N - dimethyl aniline

5) Carbylamine reaction

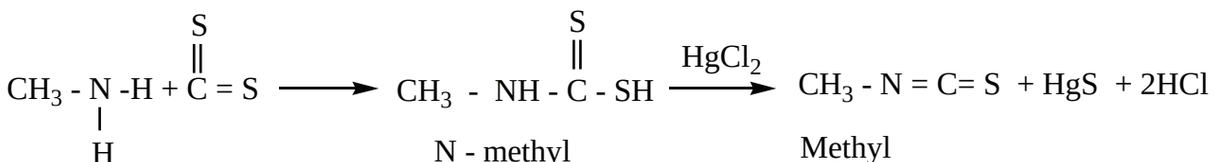
Aliphatic (or) aromatic primary amines react with chloroform and alcoholic KOH to give isocyanides (carbylamines), which has an unpleasant smell. This reaction is known as carbylamine test. This test used to identify the primary amines.

$$C_2H_5 - NH_2 + CHCl_3 + 3KOH \longrightarrow C_2H_5 - NC + 3KCl + 3H_2O$$

Ethylamine — Chloroform — Ethylisocyanide

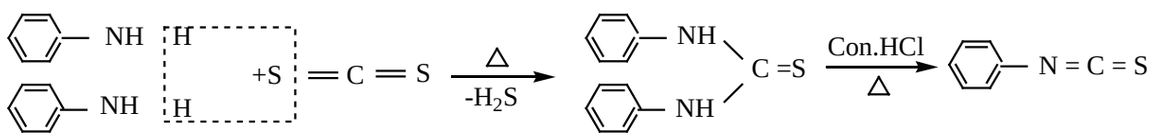
6) Mustard oil reaction

i) When primary amines are treated with carbon disulphide (CS_2), N - alkyldithio carbamic acid is formed which on subsequent treatment with $HgCl_2$, give an alkyl isothiocyanate.



Methylamine — N - methyl dithiocarbamic acid — Methyl isothiocyanate (Mustard oil smell)

ii) When aniline is treated with carbon disulphide, or heated together, S- diphenylthio urea is formed, which on boiling with strong HCl, phenyl isothiocyanate (phenyl mustard oil), is formed.



Aniline — S - diphenyl thiourea — Phenyl isothiocyanate

These reactions are known as Hofmann – Mustard oil reaction. This test is used to identify the primary amines.

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